



OPEN WA-NET: enhanced boundary-aware segmentation of skin lesions via frequency-spatial feature fusion and attention-guided edge refinement

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Skin cancer remains a major public health concern due to its high morbidity and mortality rates. While automatic segmentation techniques have improved diagnostic accuracy, they continue to face challenges such as artefacts, small lesion detection, and poor contrast between lesions and surrounding tissue. To overcome these limitations, we propose WA-NET, a novel skin lesion segmentation network that integrates a Boundary Refinement module (BRM) and an Enhanced Wavelet Transform (EWT) module. The BRM employs independent edge detection branches to enhance boundary representation, particularly in low-contrast regions. The EWT module adaptively fuses multi-scale, multi-directional sub-band features in the frequency domain to better capture texture and structural details. Furthermore, a composite loss function combining binary cross-entropy, Dice loss, and edge-supervised loss is introduced to improve both global segmentation accuracy and local boundary precision. WA-NET achieves state-of-the-art performance on three benchmark datasets—ISIC2017 (DSC: 0.9395, SE: 0.9357, ACC: 0.9573), ISIC2018 (DSC: 0.9458, SE: 0.9460, ACC: 0.9610), and PH2 (DSC: 0.9517, SE: 0.9593, ACC: 0.9638)—demonstrating strong robustness and superior boundary segmentation under challenging imaging conditions.

Keywords Skin lesion segmentation, Medical image Segmentation, Deep learning, Boundary refinement

In recent years, the rapid advancement of computer science and imaging technologies has established digital images—owing to their intuitive visual representation—as a vital medium for perceiving and interpreting information¹. Within this context, medical image segmentation has emerged as a fundamental technique in medical image analysis. It leverages computational algorithms to accurately identify regions of interest at the pixel or voxel level and classify them into specific categories². The primary objective is to delineate anatomical structures from complex backgrounds or adjacent tissues³, thereby providing objective and quantifiable image-based support for clinical diagnosis, treatment planning, and biomedical research.

Accurate feature extraction is central to medical image segmentation because segmentation quality directly influences clinical decision-making⁴. Medical images present diverse characteristics, including fine-grained local details and global structural patterns. However, boundaries are often indistinct, target regions vary considerably in shape, and foreground–background imbalance remains a common challenge⁵. In addition, medical images are susceptible to heterogeneous noise, low contrast, and significant inter-individual variability introduced by differences in imaging devices and acquisition protocols⁶. These factors necessitate the development of tailored segmentation methods that can robustly extract relevant features and address the diverse complexities inherent in medical imaging data.

Traditional methods predominantly rely on manual annotation, which is inefficient and makes it difficult to depend solely on clinicians for accurate interpretation⁷. Conventional segmentation approaches largely analyze low-level pixel features—such as color, grayscale, texture, and edges—to achieve segmentation⁸. However, when applied to complex and diverse medical images, these methods struggle to capture high-level semantic information. They mainly learn low-level features, which hinders their ability to distinguish between normal

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tissues and regions of interest, thereby limiting precise delineation of target structures. Despite these drawbacks, traditional approaches laid a solid foundation for subsequent developments^{9,10}.

With the continual progress of machine learning and deep learning technologies, segmentation methods have rapidly evolved through the integration of domain knowledge and computational advances. Deep learning models emulate the human brain's capacity to extract meaningful patterns from data through hierarchical feature learning¹¹. In recent years, deep learning has achieved remarkable success across a variety of fields.

The earliest application of neural networks to segmentation was the fully convolutional network (FCN), which replaced fully connected layers of conventional CNNs with convolutional layers¹². Although this marked a significant step forward, FCN-based results were often coarse and suffered from substantial feature loss. As a result, they were inadequate for the high precision demands of medical imaging. Building upon the FCN architecture, U-Net was introduced with optimised skip connections that alleviate spatial information loss during downsampling¹³. U-Net not only reduces the number of network parameters but also enables multi-scale feature fusion, significantly improving segmentation accuracy. It has since become a seminal model in medical image segmentation, with many variants achieving excellent performance across diverse segmentation tasks.

To further enhance multi-scale feature extraction, U-Net++¹⁴ refines the original U-Net by introducing dense skip connections at each layer, thereby improving its ability to segment complex structures and enhancing the flexibility of feature fusion. However, skip connections in U-Net architectures may still propagate irrelevant information. To address this limitation, Attention U-Net integrates an attention mechanism to dynamically emphasise informative regions while suppressing redundant ones during feature fusion¹⁵.

More recently, the Transformer architecture has attracted considerable attention due to its success in natural language processing, motivating its adoption in segmentation tasks^{16,17}. Hybrid models such as TransUNet¹⁸, Swin-UNet¹⁹, and other Transformer-enhanced variants combine U-Net's strong local feature extraction capabilities with the Transformer's global modelling strength. This integration allows the network to capture both fine-grained spatial structures and long-range semantic dependencies. Consequently, these models achieve more precise and comprehensive feature representation, thereby improving both segmentation accuracy and efficiency²⁰.

In the specific context of skin cancer analysis, boundary information plays a particularly critical role. To better exploit such information, Lin et al.²¹ proposed an edge-guided decoder network (CTO), which incorporates the Sobel edge detection operator to extract boundary features. This approach enhances edge representations on feature maps and fuses foreground–background information via an edge-guidance module, leading to more accurate boundary localisation. Collectively, these methods aim to ensure the effective transmission of salient features through hierarchical feature superposition, thereby strengthening feature representation across the entire network.

At present, most mainstream segmentation methods primarily rely on feature extraction in the spatial domain²². These approaches typically adopt encoder–decoder architectures and employ spatial-domain techniques such as multi-scale feature extraction and attention mechanisms to capture shape and structural information. However, such methods are highly sensitive to noise and struggle to capture complementary frequency-domain information. Relying solely on spatial features limits the representational capacity of segmentation models. By transforming an image from the spatial to the frequency domain, texture and frequency-related information can be effectively captured, thereby enhancing relevant features while suppressing noise and artefacts²³. Common frequency-domain feature extraction techniques include the Fourier Transform and the Wavelet Transform. While the Fourier Transform provides only global frequency information, the Wavelet Transform offers both high-frequency details and low-frequency structural contours. Furthermore, its multi-scale decomposition makes it particularly well suited to image processing tasks.

This study recognises that although existing segmentation methods have achieved notable progress, they predominantly focus on spatial-domain feature extraction. In contrast, relatively limited effort has been devoted to incorporating frequency-domain information to capture fine-grained morphological and textural characteristics. To address this gap, we propose a strategy that integrates edge information through multi-level feature fusion, thereby combining global representations with local lesion details across multiple scales to achieve more precise segmentation.

The main contributions of this work are summarised as follows:

1. We propose WA-NET, a novel segmentation model based on the classical U-Net architecture. An *Enhanced Wavelet Transform (EWT)* module is designed to integrate Haar wavelet decomposition with a channel–spatial attention mechanism. This design enables dynamic fusion of sub-band features across multiple scales and orientations in the frequency domain, effectively enhancing the model's capacity to capture edge details and texture structures.
2. We introduce a *Boundary Refinement Module (BRM)* at each stage of the encoder. This module generates edge maps via independent edge-detection branches and optimises feature fusion by incorporating an attention mechanism. Multi-scale edge maps are fused using a weighted strategy, and the final edge representation is refined through channel attention, thereby improving the model's sensitivity to boundary information.
3. We design a cross-domain fusion strategy that concatenates spatial- and frequency-domain features. A composite loss function integrating Binary Cross-Entropy (BCE), Dice loss, and edge-supervised loss is employed. The edge maps generated by the edge branches are directly involved in training through the edge-supervised loss. This strategy guides the network to learn sharper and more accurate boundaries. The effectiveness of the proposed model is validated on three public medical image datasets, demonstrating significant improvements in boundary segmentation accuracy.

Overall, the proposed WA-NET integrates frequency- and spatial-domain representations within a unified framework. The EWT module extracts multi-directional frequency features, while the channel-spatial attention mechanism enhances both high-frequency edge details and low-frequency texture cues. In parallel, the BRM generates multi-scale edge maps and improves boundary perception through attention-guided fusion. By combining edge features across resolutions, it reduces contour discontinuities, particularly for small structures. During decoding, spatial-domain backbone features and frequency-domain wavelet features are concatenated, followed by a 1×1 convolution to maximise cross-domain complementarity.

Related work

Medical image segmentation has traditionally relied on spatial-domain features. However, critical information such as texture and edges is often more salient in the frequency domain. Consequently, fusing spatial- and frequency-domain features while enhancing boundary representations has emerged as an effective strategy to improve segmentation accuracy²⁴.

Frequency-domain feature extraction

Spatial-domain information alone may fail to capture complex structural details in medical images. The wavelet transform, with its strong time-frequency localization and multi-scale decomposition capabilities, has been shown to effectively extract texture features²⁵. Integrating spatial and frequency features can therefore yield richer, more discriminative representations.

Recent studies have incorporated wavelet transforms into deep learning frameworks. Liu et al.²⁶ proposed a multi-stage wavelet CNN, replacing standard downsampling and upsampling with discrete and inverse wavelet transforms (DWT/IWT). MARIA et al.²⁷ introduced a deep adaptive wavelet network with trainable wavelet filters within a CNN. Zhao et al.²⁸ developed WRANet, a residual attention network that propagates low-frequency information during downsampling to capture high-level features while attenuating high-frequency components. More recently, Spectral U-Net²⁹ employed spectral decomposition via Wave-Block and iWave-Block modules, streamlining the fusion of wavelet components.

While these approaches demonstrate the utility of frequency-domain analysis, most employ fixed or heuristic fusion strategies for sub-band features. They lack adaptive mechanisms to dynamically weigh the importance of different frequency components according to input content.

Edge reinforcement

Blurry and indistinct boundaries remain a persistent challenge in medical image segmentation³⁰. Yan et al.³¹ proposed CCT-Net, which uses a local coding module and cross-shaped window self-attention to enhance local feature extraction while capturing long-range dependencies. CSAP-UNet³² addresses boundary blur by introducing a Boundary Enhancement Module (BEM) that processes low-level features from CNN (ResNet) and Transformer (DeiT) pathways in parallel, improving edge localization.

These studies demonstrate the value of explicit boundary modeling. However, edge refinement is often treated as a separate or late-stage process, limiting the integration of boundary information across multiple feature scales.

Encoder-decoder architectures

Enhancements to encoder-decoder frameworks, particularly U-Net, have focused on encoder feature extraction, skip connection optimization, and decoder fusion strategies³³. ARCUNet³⁴ incorporates residual convolutions and attention mechanisms to accelerate convergence and suppress irrelevant features. MRP-UNet³⁵ constructs multi-scale feature representations with residual structures and skip connections, augmented with dilated convolutions to expand receptive fields. CoTransUNet³⁶ combines fine-grained local and long-range features through a correlation-enhancement module, improving decoder reconstruction.

Recent networks further improve lesion segmentation. SGNet³⁷ adopts lightweight attention-guided designs, while ARU-Net-MD³⁸ leverages multi-scale dense connections with attention refinement. DSU-Net³⁹ integrates CNN and Transformer branches in a dual-stage U-Net to capture both local details and global context. BDFormer⁴⁰ employs boundary-aware transformers to explicitly enhance lesion edges.

Despite these advances, most architectures primarily operate within the spatial domain. Deeply integrated, parallel frequency-spatial analysis within encoder-decoder frameworks remains underexplored.

Our approach

Building upon these insights, we propose WA-NET, a frequency-spatial parallel architecture. It integrates an Enhanced Wavelet Transform (EWT) and a Boundary Refinement Module (BRM).

The EWT module performs spatial convolution and wavelet-based frequency decomposition simultaneously. It employs a channel-spatial attention mechanism to dynamically fuse multi-scale sub-band features. In parallel, the BRM generates edge maps at each encoding level, reinforcing boundary perception throughout the network.

During decoding, wavelet-domain features are fused with spatial-domain backbone features. This cross-domain interaction maximizes feature complementarity and addresses both texture representation and boundary ambiguity in skin lesion segmentation.

Methodology

This section presents the proposed WA-NET model from two perspectives: the network architecture design and the formulation of the loss function.

WA-NET is designed as a task-driven solution to the intrinsic challenges of skin lesion segmentation, which include irregular shapes, low contrast, blurred boundaries, and heterogeneous textures. Conventional U-Net variants, relying primarily on spatial-domain convolutions, often struggle with these issues, leading to under-segmentation or boundary smoothing.

To overcome these limitations, we introduce two core modules into a U-Net-based framework: the Enhanced Wavelet Transform (EWT) and the Boundary Refinement Module (BRM). The EWT leverages wavelet-domain decomposition to explicitly separate frequency components, thereby enhancing the representation of subtle boundaries and textures. The BRM provides fine-grained edge supervision to reinforce the model's sensitivity to complex lesion contours. This dual-pathway design, fusing frequency and spatial domains with explicit boundary awareness, constitutes a principled approach tailored to the problem, rather than an incremental modification.

Architecture of WA-NET

WA-NET consists of two core components: the Enhanced Wavelet Transform (EWT) module and the Boundary Refinement Module (BRM), both embedded within a U-Net-based encoder–decoder architecture.

Enhanced wavelet transform module (EWT)

Lesion boundaries in dermoscopic images often exhibit low contrast and blurred edges, which makes it challenging for conventional convolution operations in the spatial domain to effectively capture weak boundary information. Inspired by the advantages of wavelet analysis in multi-scale signal processing, we design an Enhanced Wavelet Transform (EWT) module. By performing frequency-domain decomposition to explicitly separate edge components in different directions, the EWT module enhances the representation of ambiguous boundaries. The module is inserted at each encoder stage except the final one, and integrates multi-scale wavelet decomposition with a dual attention mechanism to adaptively refine sub-band features.

1. **Multi-scale wavelet decomposition.** Given the input feature map $x \in \mathbb{R}^{B \times C \times H \times W}$, the EWT applies fixed 2×2 Haar wavelet filters to decompose it into four frequency sub-bands:

- *LL*: low-frequency in both directions, capturing global structure and contours;
- *LH*: horizontal low-frequency and vertical high-frequency components, highlighting horizontal edges;
- *HL*: horizontal high-frequency and vertical low-frequency components, highlighting vertical edges;
- *HH*: high-frequency in both directions, capturing diagonal and fine textures.

Mathematically, the discrete Haar decomposition of x is defined as:

$$LL(i, j) = \sum_{m=0}^1 \sum_{n=0}^1 h_{LL}(m, n) \cdot x(2i + m, 2j + n), \quad (1)$$

$$LH(i, j) = \sum_{m=0}^1 \sum_{n=0}^1 h_{LH}(m, n) \cdot x(2i + m, 2j + n), \quad (2)$$

$$HL(i, j) = \sum_{m=0}^1 \sum_{n=0}^1 h_{HL}(m, n) \cdot x(2i + m, 2j + n), \quad (3)$$

$$HH(i, j) = \sum_{m=0}^1 \sum_{n=0}^1 h_{HH}(m, n) \cdot x(2i + m, 2j + n), \quad (4)$$

where the Haar kernels are:

$$\begin{aligned} h_{LL} &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, & h_{LH} &= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \\ h_{HL} &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, & h_{HH} &= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \end{aligned} \quad (5)$$

Grouped convolutions with stride 2 are used for downsampling, ensuring stable and interpretable decomposition. The resulting feature map has dimension $B \times 4C \times \frac{H}{2} \times \frac{W}{2}$.

2. **Channel attention.** To model the relative importance of sub-bands across channels, global average pooling is applied to obtain channel descriptors:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F(c, i, j). \quad (6)$$

The descriptors are passed through a two-layer MLP:

$$s = \sigma(W_2 \delta(W_1 z)), \quad (7)$$

where $W_1 \in \mathbb{R}^{\frac{4C}{r} \times 4C}$ and $W_2 \in \mathbb{R}^{4C \times \frac{4C}{r}}$ with reduction ratio r , δ is ReLU, and σ is sigmoid. The learned weights s rescale the channels, selectively enhancing informative sub-band features.

Intuitively, this step assigns different levels of importance to each frequency sub-band. For example, in dermoscopic images with blurred edges, the network may assign larger weights to the LH/HL sub-bands that capture horizontal and vertical edges, while suppressing redundant low-frequency information. This dynamic weighting ensures that the most discriminative directional features are emphasized.

3. Spatial attention. On the channel-refined features F_c , spatial attention is introduced. Average and max pooling are applied along the channel dimension:

$$F_{\text{avg}}(i, j) = \frac{1}{C} \sum_{c=1}^C F_c(c, i, j), \quad F_{\text{max}}(i, j) = \max_{c \in \{1, \dots, C\}} F_c(c, i, j). \quad (8)$$

The two maps are concatenated and convolved with a 7×7 kernel:

$$M_s = \sigma(\text{Conv}_{7 \times 7}([F_{\text{avg}}; F_{\text{max}}])), \quad (9)$$

where $M_s \in \mathbb{R}^{1 \times H \times W}$ is the spatial attention map. The refined features are obtained by element-wise multiplication:

$$F_{cs}(c, i, j) = M_s(i, j) \cdot F_c(c, i, j). \quad (10)$$

Intuitively, spatial attention highlights key lesion regions within each sub-band, such as irregular contours or texture-rich areas. It simultaneously suppresses irrelevant background responses, ensuring that edge information from frequency decomposition is preserved at the most discriminative locations.

Finally, F_{cs} is evenly split along the channel dimension into four groups $\{F_{LL}, F_{LH}, F_{HL}, F_{HH}\}$, preserving directional properties of the wavelet decomposition. These are concatenated to form the enhanced feature map:

$$F_{\text{EWT}} = [F_{LL}, F_{LH}, F_{HL}, F_{HH}]. \quad (11)$$

Through this frequency-domain decomposition and dual attention-guided fusion, the EWT module adaptively emphasizes lesion boundaries and fine textures that are hard to capture in the spatial domain, significantly improving segmentation accuracy in low-contrast or artefact-prone regions.

From a medical imaging perspective, dermoscopic lesions often exhibit irregular textures and blurred boundaries under low contrast. By explicitly decomposing features into directional frequency sub-bands, EWT provides a more discriminative representation of subtle structures such as faint pigment networks and irregular edges, which are easily overlooked by purely spatial convolutions. This makes EWT particularly effective for enhancing lesion contour clarity in challenging dermoscopy cases.

Difference from CBAM. Although the design of EWT's dual attention is inspired by CBAM, there are two major differences. First, in CBAM the attention is applied directly on convolutional feature maps in the spatial domain, whereas in EWT it is specifically applied to frequency-domain sub-bands obtained by wavelet decomposition. This allows EWT to explicitly differentiate and reweight directional edge features (LH, HL, HH) and structural contours (LL). Second, instead of treating all channels equally as in CBAM, EWT reorganizes the attention-refined features back into frequency groups, preserving the interpretability of wavelet sub-bands. Thus, EWT not only learns “where to look” and “what to emphasize,” but also explicitly leverages frequency priors that are absent in conventional CBAM.

Boundary Refinement Module (BRM) The BRM is designed to enhance the network's sensitivity to lesion contours and improve segmentation accuracy under complex or ambiguous boundary conditions. It operates in three stages:

(1) Edge prediction. Shallow encoder features F_l and deep semantic features F_h are combined and passed through a multilayer perceptron (MLP) followed by a Sigmoid activation to generate an edge probability map:

$$E = \sigma(\text{MLP}(F_l, F_h)), \quad (12)$$

where $\sigma(\cdot)$ denotes the Sigmoid function.

(2) Edge-guided fusion. The predicted edge map is concatenated with the original feature map:

$$F_{\text{edge}} = \text{Concat}(F, E), \quad (13)$$

providing explicit boundary cues for subsequent refinement.

(3) Contextual refinement. F_{edge} is processed by convolutional layers, global average pooling (GAP), and an MLP to integrate local edge information with global semantic context:

$$F' = \sigma\left(\text{Conv}\left(\text{MLP}\left(\text{GAP}\left(\text{Conv}(F_{\text{edge}})\right)\right)\right)\right). \quad (14)$$

The output F' acts as a boundary-enhanced feature map, supplying contour-aware feedback to the main network.

In dermoscopic images, lesion margins are often indistinct, irregular, or fragmented due to illumination variations and surrounding skin artifacts. By jointly modeling shallow edge cues and high-level semantic context, BRM avoids

the over-smoothing issue of conventional encoder–decoder models and produces sharper, more consistent lesion boundaries, especially in cases with fuzzy or discontinuous contours.

Overall architecture

An overview of the WA-NET architecture is shown in Fig. 1. The network follows a symmetrical encoder–decoder structure.

The encoder comprises five stages, each containing two 3×3 convolutional layers with ReLU activation. The first four stages use 2×2 max pooling for downsampling. At each stage, features are refined by BRMs, generating an edge map via a 3×3 convolution and sigmoid activation. The edge map is concatenated with the original feature map and further refined through a 3×3 convolution followed by channel–spatial attention.

The refined features from the first four encoder stages are passed through EWT modules. Haar wavelet decomposition extracts four frequency-domain sub-bands, which are weighted using channel–spatial attention. The sub-band features are aligned with corresponding encoder features via a feature alignment module. A dynamic fusion module then adaptively combines the features using a 1×1 convolution with learnable weights.

The decoder mirrors the encoder. Each stage performs bilinear upsampling of the feature map, concatenates it with the corresponding fused encoder features via skip connections, reduces channel dimensionality using a 1×1 convolution, and refines the result with two 3×3 convolutions.

For boundary processing, edge maps from all five BRMs are upsampled to the input resolution via bilinear interpolation. Each map is refined using 3×3 convolutions and fused using softmax-normalized learnable weights. Efficient Channel Attention generates the final edge map, which serves as both an auxiliary output and a regularization signal.

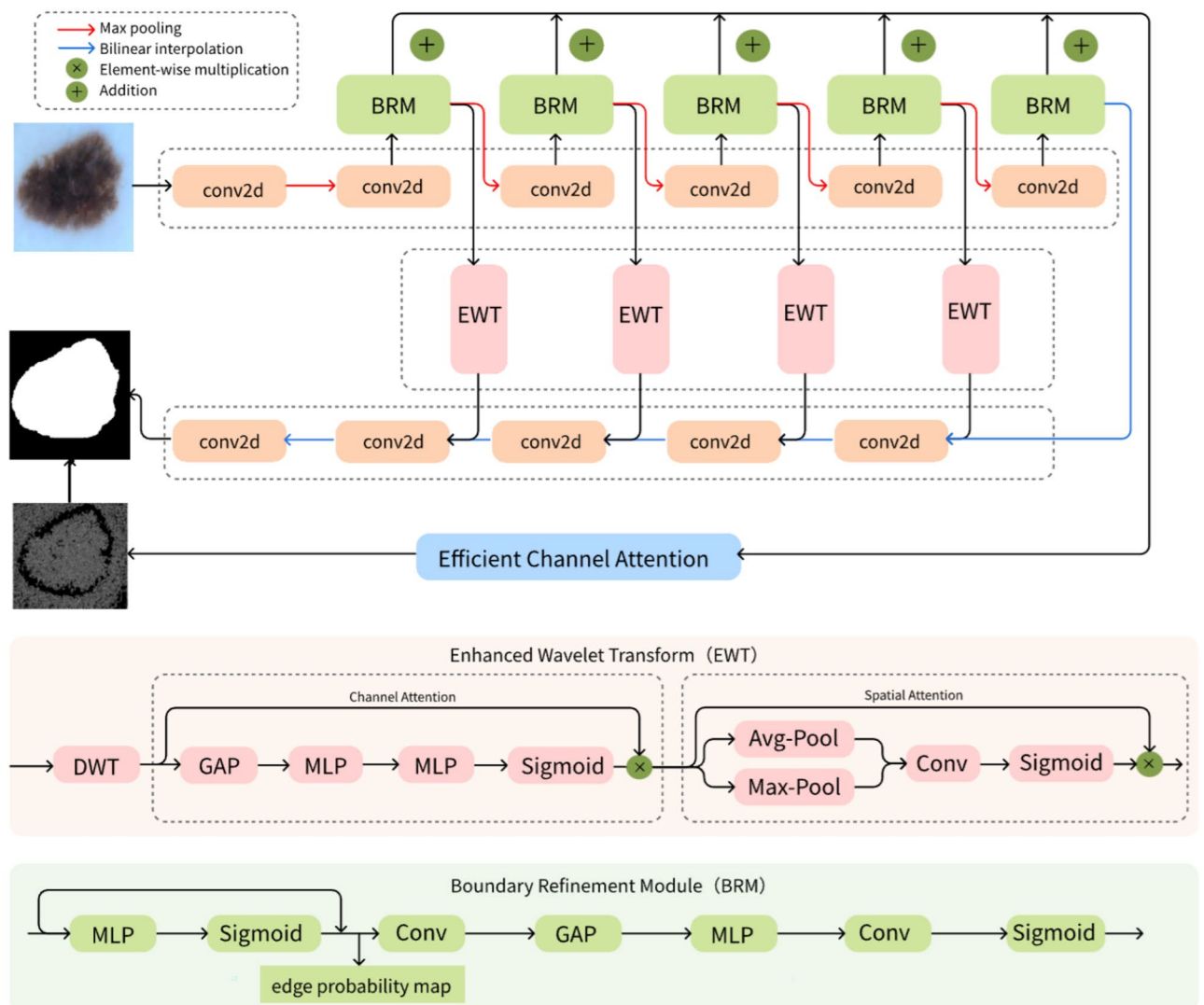


Fig. 1. Overall structure of WA-NET.

During training, binary edge labels are generated by applying the Sobel operator to ground-truth masks. These labels supervise the BRM-predicted edge maps via binary cross-entropy loss. The edge branch is a lightweight convolutional sub-network trained to produce soft edge predictions that align with anatomical structures.

Pseudo-code of core modules

To improve reproducibility and clarity, we provide the pseudo-code for the Enhanced Wavelet Transform (EWT) and the Boundary Refinement Module (BRM) in Algorithm 1 and Algorithm 2.

Require: Input feature map $X \in \mathbb{R}^{B \times C \times H \times W}$

Ensure: Enhanced feature map $\hat{F} \in \mathbb{R}^{B \times 4C \times H/2 \times W/2}$

- 1: **Wavelet decomposition:** Apply fixed Haar filters to decompose X into four sub-bands $\{LL, LH, HL, HH\}$.
 - 2: **Channel attention:** Compute global descriptors, pass through MLP, and reweight channels adaptively.
 - 3: **Spatial attention:** Generate attention map from pooled features, refine spatially important regions.
 - 4: **Reorganization:** Concatenate refined sub-bands to form the final representation $\hat{F}$.
 - 5: **return** $\hat{F}$
-

Algorithm 1. Enhanced Wavelet Transform (EWT) with Dual Attention.

Require: Input feature map $X \in \mathbb{R}^{B \times C \times H \times W}$

Ensure: Refined feature map X_{ref} , boundary map E

- 1: **Edge prediction:** Generate boundary probability map E from shallow and deep features.
 - 2: **Edge-guided fusion:** Concatenate E with original features X to obtain X_{edge} .
 - 3: **Feature refinement:** Apply attention mechanism to X_{edge} to enhance boundary-sensitive regions.
 - 4: **return** X_{ref}, E
-

Algorithm 2. Boundary Refinement Module (BRM).

We evaluated the computational complexity of WA-NET at both module and model levels.

- **EWT:** 4.19 MFLOPs with negligible trainable parameters due to depthwise convolution design.
- **BRM:** 14.79K parameters and 239.63 MFLOPs.
- **Full WA-NET:** 11.6M parameters and 10.52 GFLOPs, measured using fvcare. Element-wise operations (e.g., sigmoid, mean, add) are excluded but negligible compared to convolutions.

Loss function

To simultaneously enhance pixel-wise classification accuracy, boundary sharpness, and structural integrity, we design a composite loss function that combines segmentation loss, edge supervision loss, and region similarity loss.

Binary Cross-Entropy Loss. As a standard pixel-level loss, BCE quantifies the discrepancy between predicted probabilities and binary ground truth labels:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)] \quad (15)$$

where $y_i \in \{0, 1\}$ is the ground truth label and $p_i \in [0, 1]$ is the predicted probability.

Dice Loss. To mitigate foreground-background imbalance, we incorporate Dice loss, which measures overlap:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N p_i y_i}{\sum_{i=1}^N p_i + \sum_{i=1}^N y_i} \quad (16)$$

Edge Supervision Loss. An auxiliary edge supervision branch is introduced. Ground-truth edge maps are generated using the Sobel operator on binary masks. The predicted edge map is supervised by a binary cross-entropy loss:

$$\mathcal{L}_{\text{Edge}} = -\frac{1}{N} \sum_{i=1}^N [y_i^e \log p_i^e + (1 - y_i^e) \log(1 - p_i^e)] \quad (17)$$

Total Loss. The overall loss combines the above components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \lambda \mathcal{L}_{\text{Edge}} \quad (18)$$

where λ is a balancing coefficient that controls the importance of boundary supervision.

Choice of hyperparameter λ

Experiments demonstrate that setting $\lambda = 0.7$ achieves the best performance across multiple evaluation metrics (Fig. 2). This choice is based on the following observations: edge pixels typically account for only 1–3% of the total image pixels, and a higher λ helps to mitigate this class imbalance. Moreover, accurate boundary detection is critical for clinical tasks such as asymmetry analysis.

Experimental setup and implementation

To comprehensively evaluate the effectiveness and robustness of WA-NET, segmentation experiments were conducted on three publicly available skin lesion datasets: ISIC 2017, ISIC 2018, and PH2. These datasets encompass diverse imaging conditions, lesion types, and visual characteristics, enabling a thorough assessment of the model's generalization capability.

The proposed WA-NET model comprises five encoding layers. All layers except the last incorporate the Enhanced Wavelet Transform (EWT) module. The EWT performs multi-scale decomposition using fixed Haar wavelet kernels, separating low-frequency (LL) structural information from high-frequency (LH, HL, HH) texture details. A channel-spatial attention mechanism then adaptively emphasizes salient frequency components while suppressing redundant information. This enhances the model's sensitivity to critical texture features. After wavelet decomposition, the number of channels in each feature map quadruples. These frequency-domain features are fused with backbone features during decoding, significantly improving multi-scale representation capability.

In addition, a Boundary Refinement Module (BRM) is embedded at the output of each encoder layer. The module first extracts edge-related features to generate boundary probability maps, which are then concatenated with the original feature maps and further enhanced using a channel-spatial attention mechanism. During decoding, edge features are integrated via a multi-level Edge-guided fusion to guide segmentation predictions, resulting in clearer and more coherent lesion boundaries. In summary, the EWT module focuses on frequency-domain texture modeling, while the BRM strengthens boundary perception; their synergistic operation equips WA-NET with superior modeling capability for images with blurred boundaries and complex textures.

A multi-stage data augmentation strategy was employed to maximize sample diversity and improve robustness against complex transformations and clinical imaging variations. The augmentation pipeline includes:

1. **Synchronous geometric transformations:** applied to both images and masks, including random horizontal/vertical flips, rotations (0° , 90° , 180° , 270°), and random cropping (retaining 80%–100% of the area) to enhance rotational invariance and multi-scale feature learning.

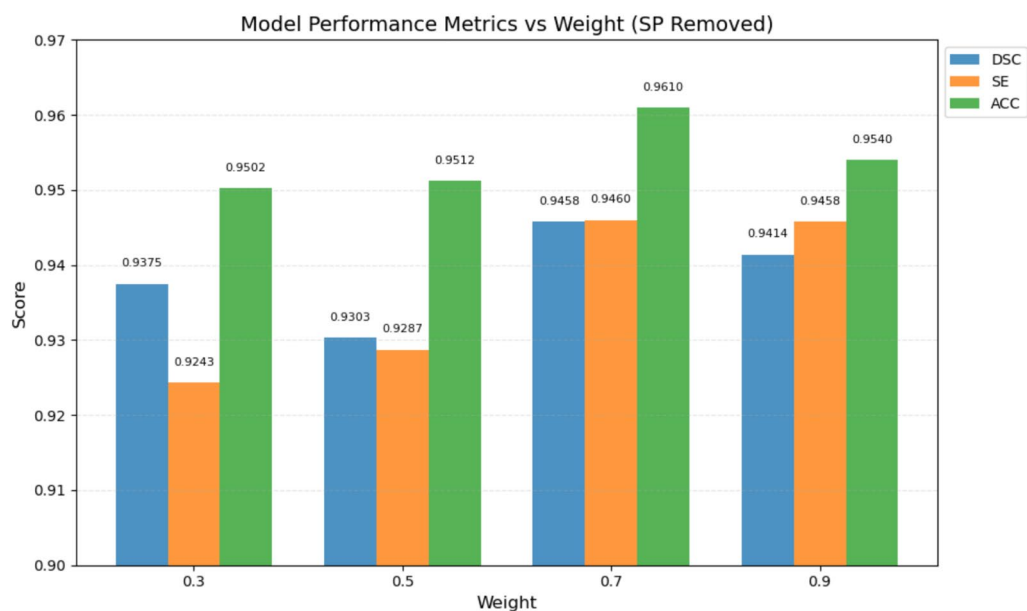


Fig. 2. Performance metrics versus different λ values.

- 2. **Image-specific augmentations:** applied only to input images, such as random contrast adjustment (70%–130%) and Gaussian blurring ($\sigma \in [0.5, 1.5]$), simulating diverse imaging conditions and improving robustness against low resolution and noise.
- 3. **Resizing:** all augmented samples are resized to 256×256 pixels to ensure consistent input dimensions and facilitate cross-dataset generalization.

Experiments were implemented in PyTorch and accelerated using an NVIDIA GeForce RTX 4060 GPU. The Adam optimizer with an initial learning rate of 1×10^{-4} and a batch size of 8 was used. Mixed-precision training improved efficiency and stability. A multi-loss joint optimization strategy was adopted, with the edge loss weight set to 0.7. Training was conducted for 100 epochs, with five-fold cross-validation on each dataset to enhance robustness.

Segmentation performance was evaluated using Dice coefficient, Accuracy, and Sensitivity. These metrics provide a comprehensive assessment of WA-NET and a solid basis for comparison with other state-of-the-art methods.

Comparative results

For baseline models whose official implementations and reported metrics are publicly available, we directly cite their results from the corresponding original publications: U-Net++¹⁴, Attention U-Net¹⁵, and TransUNet¹⁸ as reported in the BAG-Net framework⁴¹; TMAHUNet³⁴ and MHorUNet³⁵ from the multiscale input fusion U-Net study^{35,42}; and SGNet³⁷, ARU-Net-MD³⁸, DSU-Net³⁹, and BDFormer⁴⁰ from their respective original papers. The proposed WA-NET was trained and evaluated using the same datasets and metrics, ensuring that all reported comparisons are meaningful.

As shown in Table 1, WA-NET achieves a DSC of 0.9395, SE of 0.9357, and ACC of 0.9573 on the ISIC 2017 dataset, outperforming the next best method, TMAHUNet (DSC = 0.8973, SE = 0.8638), by 4.22% in DSC and 8.33% in SE, respectively. Compared to U-Net++ and Attention U-Net, WA-NET improves DSC by 6.79% and 6.18%, demonstrating its superior capability in boundary and texture modeling.

On the ISIC 2018 dataset, which exhibits greater lesion variability, WA-NET attains DSC = 0.9458, SE = 0.9460, and ACC = 0.9610. This represents an improvement of 6.14% in DSC over TMAHUNet and 6.39% in SE over U-Net++, highlighting the robustness of WA-NET in capturing complex lesion morphologies. Although ARU-Net-MD achieves a marginally higher SE (0.9494), WA-NET provides the best overall balance across all metrics.

On the PH2 dataset, WA-NET achieves competitive results with DSC = 0.9517, SE = 0.9593, and ACC = 0.9638. It is worth noting that DSU-Net shows strong performance on this dataset; however, WA-NET achieves state-of-the-art performance on the larger and more challenging ISIC 2017 and ISIC 2018 benchmarks. This consistent top-tier performance across datasets of varying scales and complexities validates the effectiveness, robustness, and superior generalization capability of WA-NET.

These results collectively demonstrate that WA-NET effectively captures frequency–spatial correlations and edge-aware representations, leading to precise delineation of skin lesions across diverse conditions. WA-NET thus represents a task-specific design tailored for skin lesion segmentation, rather than a mere incremental improvement over conventional U-Net variants.

Figure 3 provides a qualitative comparison of the segmentation outputs for various models on ISIC 2018. The red contours denote the ground truth, and the blue contours indicate the predicted boundaries. As evident, WA-NET’s predictions align closely with the ground truth, exhibiting minimal discrepancy. In contrast, other models show varying degrees of misalignment, particularly near lesion boundaries. WA-NET demonstrates superior edge adherence and produces cleaner and more accurate segmentations, especially in challenging cases with irregular lesion borders or low contrast. Figure 4 shows the training and validation loss curves for the three datasets.

Methods	ISIC 2017			ISIC 2018			PH2		
	DSC	SE	ACC	DSC	SE	ACC	DSC	SE	ACC
U-Net++ ^{14†}	0.8716	0.8750	0.9527	0.8730	0.8821	0.9427	0.9412	0.9238	0.9627
Att U-Net ^{15†}	0.8777	0.9224	0.9513	0.8731	0.8483	0.9313	0.9082	0.9202	0.9568
TransUNet ^{18†}	0.8123	0.8263	0.9207	0.8499	0.8578	0.9452	0.9333	0.9073	0.9581
TMAHUNet ^{34†}	0.8973	0.8638	0.9081	0.8844	0.8446	0.9212	0.9174	0.8563	0.9383
MHorUNet ^{35†}	0.8555	0.8783	0.8858	0.8754	0.8759	0.8754	0.8683	0.8962	0.8793
SGNet ^{37†}	0.9102	0.9111	0.9598	0.9045	0.9161	0.9573	0.9142	0.9484	0.9426
ARU-Net-MD ^{38†}	0.8906	0.9303	0.9522	0.9234	0.9494	0.9535	0.9265	0.9161	0.9656
DSU-Net ^{39†}	0.8650	0.8470	0.9404	0.9004	0.9222	0.9431	0.9604	0.9627	0.9780
BDFormer ^{40†}	0.8955	0.8938	0.9600	0.9022	0.8938	0.9602	0.9266	0.9515	0.9514
Proposed	0.9395	0.9357	0.9573	0.9458	0.9460	0.9610	0.9517	0.9593	0.9638

Table 1. Comparison of segmentation performance across ISIC 2017, ISIC 2018, and PH2 datasets. †Results are directly cited from the corresponding original publications. All cited studies and our experiments were conducted on the ISIC 2017, ISIC 2018, and PH2 datasets under comparable evaluation settings

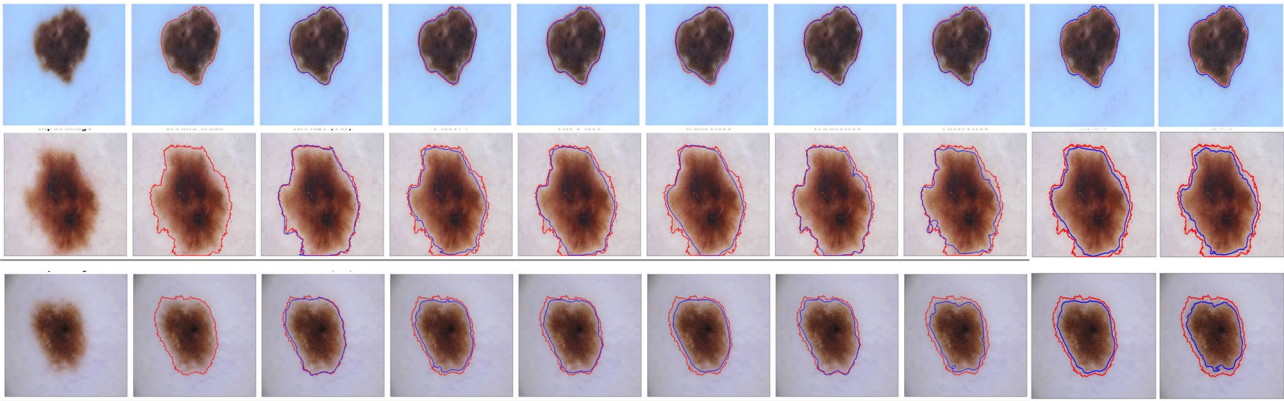


Fig. 3. Visual comparison of segmentation results on the ISIC 2018 dataset. Each row shows (from left to right): input image, ground truth (red contour), and predicted results from WA-NET, U-Net++, Attention U-Net, TransUNet, TMAHUNet, MHorUNet, ARU-Net-MD, and BDFormer (all shown with blue contours).

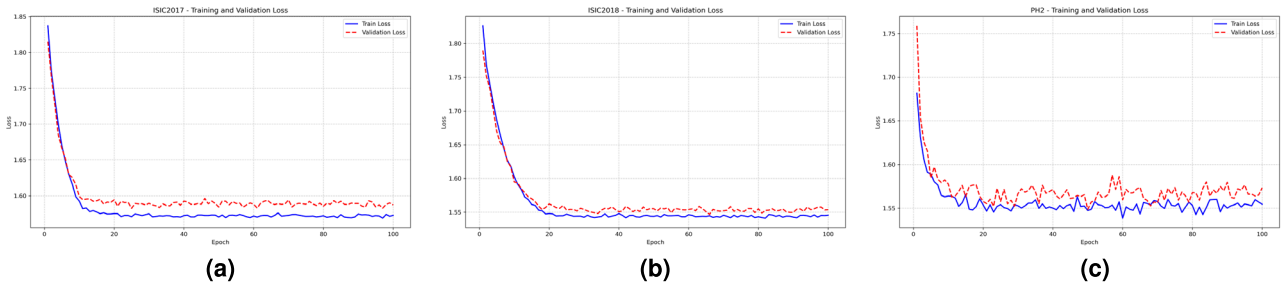


Fig. 4. Training and validation loss curves for the three datasets. The steadily decreasing trends indicate good convergence behaviour and model stability. (a) ISIC 2017, (b) ISIC 2018, (c) PH2.

Model Components				ISIC 2017			ISIC 2018			PH2		
Base	EWT	BRM	Edge	DSC	SE	ACC	DSC	SE	ACC	DSC	SE	ACC
✓	✓			0.9153	0.9145	0.9414	0.9244	0.9279	0.9417	0.9371	0.9437	0.9444
✓		✓		0.9295	0.9331	0.9534	0.9301	0.9342	0.9512	0.9417	0.9488	0.9506
✓	✓	✓		0.9318	0.9316	0.9523	0.9367	0.9352	0.9529	0.9461	0.9491	0.9522
✓	✓	✓	✓	0.9395	0.9357	0.9573	0.9458	0.9460	0.9610	0.9517	0.9593	0.9638

Table 2. Ablation Study: Performance impact of each component in WA-NET. Base = baseline architecture, EWT = Enhanced Wavelet Transform, BRM = Boundary Refinement Module, Edge = Edge Loss function

Ablation study

To evaluate the individual contributions of each module within WA-NET, we conducted an ablation study using three model variants: Base+EWT, Base+BRM, and Base+EWT+BRM (without edge loss). Table 2 summarises the quantitative results across all datasets using DSC, SE, and ACC metrics. In the ablation study, the ‘Base’ refers to the backbone architecture derived from the classical U-Net. Specifically, it retains the fundamental encoder-decoder structure with five encoding stages (each consisting of two 3x3 convolutional layers followed by 2x2 max pooling) and corresponding decoder stages (using bilinear upsampling and feature concatenation), along with the standard skip connections. Crucially, both the Enhanced Wavelet Transform (EWT) modules and the Boundary Refinement Modules (BRM) are removed from this baseline configuration. All other components, such as the final convolutional layers for output and the basic training setup (data augmentation, optimizer, learning rate, etc.), remain consistent with the full WA-NET model.

As shown in Table 2, the full WA-NET model outperforms all ablated variants. The inclusion of both EWT and BRM modules leads to significant improvements over the base model. Notably, adding BRM alone boosts accuracy more than EWT alone, underscoring the importance of accurate boundary learning. Furthermore, incorporating the edge-aware loss function delivers the highest performance across all datasets, reinforcing the role of edge supervision in enhancing segmentation precision.

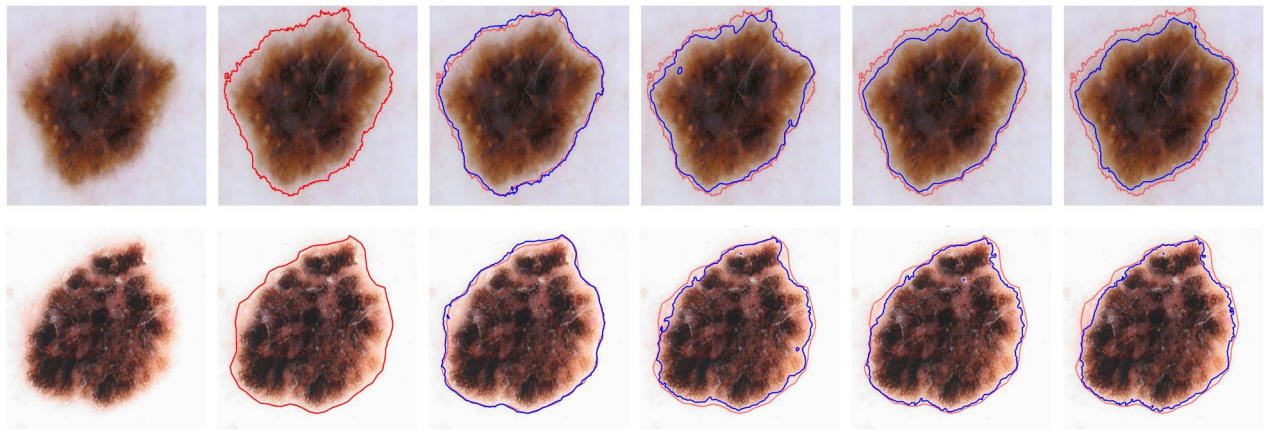


Fig. 5. Qualitative results from the ablation study on the ISIC 2018 dataset. From left to right: input image, ground truth (in red), WA-NET full model, Base+BRM, Base+EWT, and Base+EWT+BRM (without edge loss). The predictions of all models are outlined in blue, while the ground truth contours are shown in red. WA-NET yields the most accurate and boundary-aligned segmentation results.

To further validate the visual impact of each module, Fig. 5 provides a qualitative comparison on the ISIC 2018 dataset, including the input image, ground truth, and predictions from the Full model, Base+BRM, and Base+EWT+BRM (without edge loss), Base configurations. In these visualisations, the red contour denotes the ground truth, while the blue contour indicates the predicted segmentation.

Conclusion

In this paper, we propose WA-NET, a novel medical image segmentation network that effectively integrates frequency-domain and spatial-domain features. WA-NET leverages Haar wavelet decomposition to extract multi-scale, multi-directional sub-band features and employs an attention mechanism to adaptively weight the fusion of these frequency components, thereby capturing their relative importance across different bands. A Boundary Refinement Module is embedded within each encoder layer to enhance edge sensitivity through multi-stage processing, including edge feature extraction, boundary map prediction, feature fusion, and attention-based enhancement. The generated edge maps not only guide the training process but also directly improve the accuracy of final segmentation boundaries.

Regarding the loss function, we design a composite objective combining binary cross-entropy, Dice loss to mitigate class imbalance, and edge supervision loss, which collectively balances global segmentation accuracy with local boundary precision and enhances structural segmentation capability.

Importantly, the proposed WA-NET is not intended as a simple incremental modification of U-Net, but as a task-driven design tailored to the intrinsic challenges of skin lesion segmentation. By explicitly addressing issues such as irregular lesion morphology, low contrast, and blurred or ambiguous boundaries, WA-NET demonstrates that frequency-spatial feature fusion combined with boundary refinement provides a principled and effective solution for enhancing segmentation accuracy in complex medical imaging scenarios.

Data availability

The datasets utilized in this study are all publicly accessible: • ISIC 2017 Challenge dataset: <https://challenge.isi-c-archive.com>. • ISIC 2018 Challenge dataset: <https://challenge2018.isic-archive.com>. • PH2 dermoscopic image dataset: <https://www.kaggle.com/datasets/athina123/ph2dataset>.

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References

1. Xiao, H., Li, L., Liu, Q., Zhu, X. & Zhang, Q. Transformers in medical image segmentation: A review. *Biomedical Signal Processing and Control* **84**, 104791. <https://doi.org/10.1016/j.bspc.2023.104791> (2023).
2. Ates, G. C., Mohan, P. N. & Celik, E. Dual cross-attention for medical image segmentation. *Engineering Applications of Artificial Intelligence* **126**, 107139. <https://doi.org/10.1016/j.engappai.2023.107139> (2023).
3. Munia, A. A. et al. Attention-guided hierarchical fusion u-net for uncertainty-driven medical image segmentation. *Information Fusion* 102719–102719, <https://doi.org/10.1016/j.inffus.2024.102719> (2024).
4. Wu, R., Liu, Y., Liang, P. & Chang, Q. H-vmmnet: High-order vision mamba unet for medical image segmentation. *Neurocomputing* **624**, 129447. <https://doi.org/10.1016/j.neucom.2025.129447> (2025).
5. Meng, W., Liu, S. & Wang, H. Afc-unet: Attention-fused full-scale cnn-transformer unet for medical image segmentation. *Biomedical Signal Processing and Control* **99**, 106839. <https://doi.org/10.1016/j.bspc.2024.106839> (2025).
6. Zhang, Y. et al. Low-contrast medical image segmentation via transformer and boundary perception. *IEEE Transactions on Emerging Topics in Computational Intelligence* **8**, 2297–2309. <https://doi.org/10.1109/tetci.2024.3353624> (2024).
7. Bai, R. & Zhou, M. Sl-hardnet: Skin lesion segmentation with hardnet. *Frontiers in Bioengineering and Biotechnology* **10**, 1028690. <https://doi.org/10.3389/fbioe.2022.1028690> (2023).

8. Bohmrah, M. K. & Kaur, H. Advanced hybridization and optimization of DNNs for medical imaging: A survey on disease detection techniques. *Artificial Intelligence Review* **58**, 122. <https://doi.org/10.1007/s10462-024-11049-x> (2025).
9. Maurya, S. et al. A review on recent developments in cancer detection using machine learning and deep learning models. *Biomedical Signal Processing and Control* **80**, 104398. <https://doi.org/10.1016/j.bspc.2023.104398> (2023).
10. Azad, R., Khosravi, N., Dehghanmanshadi, M., Cohen-Adad, J. & Merhof, D. Medical image segmentation on mri images with missing modalities: A review. *arXiv preprint arXiv:2203.06217* (2022).
11. Balraj, K. et al. MADR-Net: Multi-scale attention dilated residual network for medical image segmentation. *Scientific Reports* **14**, 12699. <https://doi.org/10.1038/s41598-024-63538-2> (2024).
12. Yang, X. et al. Automatic pixel-level crack detection and measurement using fully convolutional network. *Computer-Aided Civil and Infrastructure Engineering* **33**, 1090–1109. <https://doi.org/10.1111/mice.12412> (2018).
13. Uzun, Y. & Bilgin, M. Medical image enhancement using war strategy optimization algorithm. *Biomedical Signal Processing and Control* **106**, 107740. <https://doi.org/10.1016/j.bspc.2024.107740> (2025).
14. Zhou, Z., Siddiquee, M. M.R., Tajbakhsh, N. & Liang, J. Unet++: A nested u-net architecture for medical image segmentation. In Stoyanov, D. & et al. (eds.) *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support. DLMIA/ML-CDS 2018*, vol. 11045 of *Lecture Notes in Computer Science*, https://doi.org/10.1007/978-3-030-00889-5_1 (Springer, 2018).
15. Oktay, O. et al. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999* (2018). Preprint.
16. Liu, X., Hu, Y. & Chen, J. Hybrid cnn-transformer model for medical image segmentation with pyramid convolution and multi-layer perceptron. *Biomedical Signal Processing and Control* **86**, 105331. <https://doi.org/10.1016/j.bspc.2023.105331> (2023).
17. Xiong, Y. et al. Marsformer: Martian rock semantic segmentation with transformer. *IEEE Transactions on Geoscience and Remote Sensing* **61**, 1–15. <https://doi.org/10.1109/TGRS.2023.3302649> (2023).
18. Chen, J. et al. Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306* (2021). Preprint.
19. Cao, H. et al. Swin-unet: Unet-like pure transformer for medical image segmentation. In *European Conference on Computer Vision (ECCV)*, 205–218, https://doi.org/10.1007/978-3-031-25066-8_9 (Springer, 2022).
20. Li, X. et al. MSDUNet: A model based on feature multi-scale and dual-input dynamic enhancement for skin lesion segmentation. *IEEE Transactions on Medical Imaging* <https://doi.org/10.1109/tmi.2025.3549011> (2025). Early Access.
21. Lin, Y. et al. Rethinking boundary detection in deep learning models for medical image segmentation. In Frangi, A., de Bruijne, M., Wassermann, D. & Navab, N. (eds.) *Information Processing in Medical Imaging (IPMI)*, vol. 13939 of *Lecture Notes in Computer Science*, 723–735, https://doi.org/10.1007/978-3-031-34048-2_56 (Springer, Cham, 2023).
22. Han, Q. et al. Ehdfl: Evolutionary hybrid domain feature learning based on windowed fast fourier convolution pyramid for medical image classification. *Computers in Biology and Medicine* **152**, 106353. <https://doi.org/10.1016/j.combiomed.2022.106353> (2023).
23. Li, Y., Liu, Z., Yang, J. & Zhang, H. Wavelet transform feature enhancement for semantic segmentation of remote sensing images. *Remote Sensing* **15**, 5644. <https://doi.org/10.3390/rs15245644> (2023).
24. Kumar, R. & Priyadarshi, R. Denoising and segmentation in medical image analysis: a comprehensive survey of machine learning and deep learning approaches. *Multimedia Tools and Applications* **84**, 10817–10875. <https://doi.org/10.1007/s11042-024-19313-6> (2025).
25. Zeng, Y. et al. Wet-unet: Wavelet integrated efficient transformer networks for nasopharyngeal carcinoma tumor segmentation. *Science Progress* **107**, <https://doi.org/10.1177/00368504241232537> (2024).
26. Gao, G., Lai, H. & Jia, Z. Munet++: Multilevel wavelet nested unet++ demoiréing residual network. *Displays* **83**, 102741. <https://doi.org/10.1016/j.displa.2024.102741> (2024).
27. Rodriguez, M. X.B. et al. Deep adaptive wavelet network. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 3100–3108, <https://doi.org/10.1109/WACV45572.2020.9093580> (2020).
28. Zhao, Y., Wang, S., Zhang, Y., Qiao, S. & Zhang, M. WRANet: Wavelet integrated residual attention u-net network for medical image segmentation. *Complex & Intelligent Systems* **9**, 6971–6983. <https://doi.org/10.1007/s40747-023-01119-y> (2023).
29. Peng, Y., Sonka, M. & Chen, D.Z. Spectral u-net: Enhancing medical image segmentation via spectral decomposition. *arXiv preprint arXiv:2409.09216* (2024). Preprint, Accessed: September 13, 2024, 2409.09216.
30. Wang, W. et al. An improved boundary-aware u-net for ore image semantic segmentation. *Sensors* **21**, 2615. <https://doi.org/10.3390/s21082615> (2021).
31. Yan, Y. et al. Cct-unet: a u-shaped network based on convolution coupled transformer for segmentation of peripheral and transition zones in prostate mri. *IEEE Journal of Biomedical and Health Informatics* **27**, 5963–5974. <https://doi.org/10.1109/JBHI.2023.3289913> (2023).
32. Fan, X., Zhou, J., Jiang, X., Xin, M. & Hou, L. Csap-unet: Convolution and self-attention paralleling network for medical image segmentation with edge enhancement. *Computers in Biology and Medicine* **172**, 108265. <https://doi.org/10.1016/j.combiomed.2024.108265> (2024).
33. Wu, R. et al. Hsh-unet: Hybrid selective high order interactive u-shaped model for automated skin lesion segmentation. *Computers in Biology and Medicine* **168**, 107798 (2024).
34. Soni, T. et al. Arcunet: enhancing skin lesion segmentation with residual convolutions and attention mechanisms for improved accuracy and robustness. *Scientific Reports* **15**, <https://doi.org/10.1038/s41598-025-94380-9> (2025).
35. Liu, Z., Hu, J., Gong, X. & Li, F. Skin lesion segmentation with a multiscale input fusion u-net incorporating res2-se and pyramid dilated convolution. *Scientific Reports* **15**, <https://doi.org/10.1038/s41598-025-92447-1> (2025).
36. Gao, Y., Zhang, S., Shi, L., Zhao, G. & Shi, Y. Collaborative transformer u-shaped network for medical image segmentation. *Applied Soft Computing* **112841**, <https://doi.org/10.1016/j.asoc.2025.112841> (2025).
37. Yun, H. et al. Sgnet: A structure-guided network with dual-domain boundary enhancement and semantic fusion for skin lesion segmentation. *Sensors* **25**, 4652. <https://doi.org/10.3390/s25154652> (2025).
38. Kaur, R. & Kaur, S. Aru-net-md: Automatic skin lesion segmentation using attention residual u-net with improved encoder-decoder architecture. *Multimedia Tools and Applications* **84**, 4315–4341. <https://doi.org/10.1007/s11042-024-18895-5> (2025).
39. Zhong, L., Li, T., Cui, M. et al. Dsu-net: Dual-stage u-net based on cnn and transformer for skin lesion segmentation. *Biomedical Signal Processing and Control* **100**, <https://doi.org/10.1016/j.bspc.2024.107090> (2025).
40. Ji, Z., Ye, Y. & Ma, X. Bdformer: Boundary-aware dual-decoder transformer for skin lesion segmentation. *Artificial Intelligence in Medicine* **162**, 103079. <https://doi.org/10.1016/j.artmed.2025.103079> (2025).
41. Lei, Y. et al. Bag-net: A novel architecture for enhanced medical image segmentation with global context attention and boundary self-attention. *Symmetry* **17**, 531. <https://doi.org/10.3390/sym17040531> (2025).
42. Wu, R. et al. Mhorunet: High-order spatial interaction unet for skin lesion segmentation. *Biomed. Signal Process. Control* **88**, 105517. <https://doi.org/10.1016/j.bspc.2023.105517> (2024).

Author contributions

G.H. designed the model and performed experiments; W.Z. supervised the project; X.L. and Q.L. contributed to data preprocessing and analysis; all authors wrote and reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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